

m00nwalk

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Decode this message from the moon.

Hints

- How did pictures from the moon landing get sent back to Earth?
 - What is the CMU mascot? That might help select a RX option.
-

Tools Used

- `file` – Determine file type
 - `strings` – Extract printable character sequences
 - `exiftool` – Read audio metadata
 - **Audacity** – Audio analysis software (spectrogram view)
 - **SSTV (Slow Scan Television)** – Decode images from audio signals
 - **QSSTV** – GUI alternative for SSTV decoding
-

Complete Solution

Step 1: Download and Prepare

Download the audio file named `message.wav` from the challenge page.

Step 2: Basic File Inspection

First, verify the file type:

```
file message.wav
```

Expected output:

```
message.wav: RIFF (little-endian) data, WAVE audio, Microsoft PCM, 16 bit, mono 48000 Hz
```

The file is a standard **WAV audio file** – mono, 16-bit, 48 kHz sample rate.

Step 3: Search for Visible Text

Check if the flag appears as plain text within the audio file:

```
strings message.wav | grep -i "picoCTF"
```

Output: (no results)

The flag isn't directly embedded as text.

Step 4: Inspect Metadata

```
exiftool message.wav
```

Output:

```
ExifTool Version Number      : 13.50
File Name                    : message.wav
Directory                   : .
File Size                    : 11 MB
...
File Type                   : WAV
File Type Extension          : wav
MIME Type                   : audio/x-wav
Encoding                    : Microsoft PCM
```

Num Channels	: 1
Sample Rate	: 48000
Avg Bytes Per Sec	: 96000
Bits Per Sample	: 16
Duration	: 0:01:55

Nothing suspicious in the metadata.

Step 5: Listen to the Audio

Play the audio file:

```
# Linux
```

```
aplay message.wav
```

```
# macOS
```

```
afplay message.wav
```

```
# Windows (PowerShell)
```

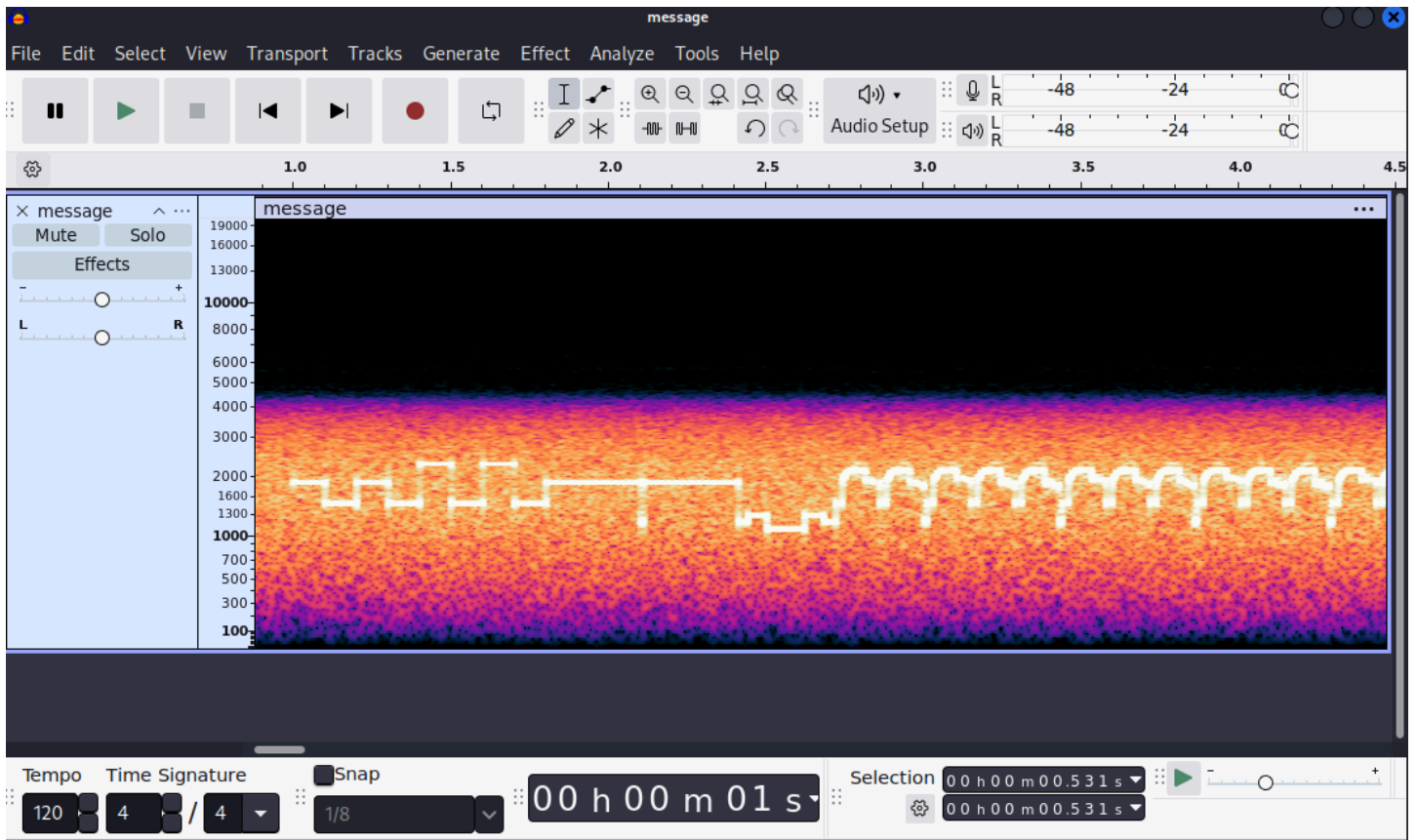
```
(New-Object Media.SoundPlayer "message.wav").PlaySync()
```

The audio sounds like a series of beeps, tones, or static – not human speech or music. This suggests it might be encoded data.

Step 6: Visual Analysis with Audacity

Open the file in **Audacity** (free audio editor):

1. Launch Audacity
2. File → Import → Audio → Select `message.wav`



Now change the view to **Spectrogram**:

- Click on the track title dropdown (left of the waveform)
- Select **Spectrogram**

The spectrogram reveals a **circuit-like pattern** or **image**! This is characteristic of **Slow Scan Television (SSTV)** – a technique used to transmit images over radio frequencies.

Historical context: The Apollo missions used SSTV to transmit images from the moon back to Earth. This challenge name "m00nwalk" is a direct reference.

Step 7: Decode with SSTV (Command Line)

SSTV is a protocol for transmitting images via audio. We need to decode the audio back into an image.

Install SSTV decoder (Python version):

```
git clone https://github.com/colaclanth/sstv.git
cd sstv
python setup.py install
```

Decode the audio:

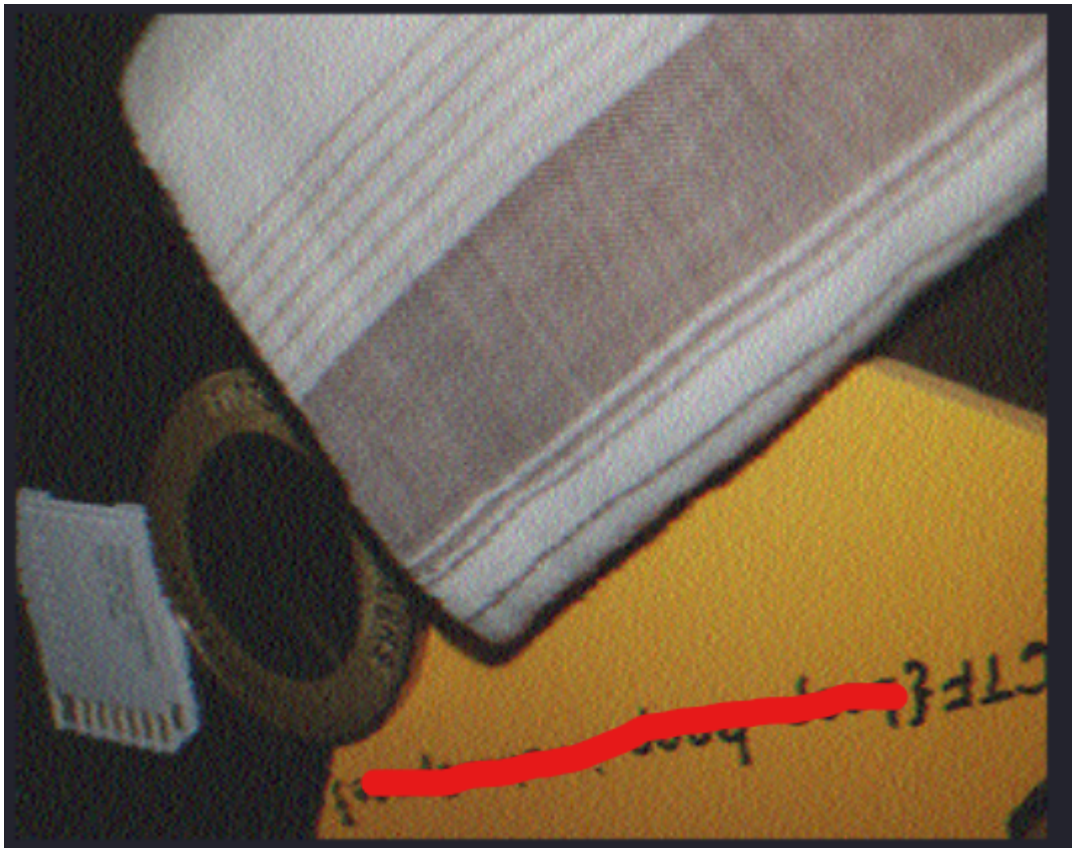
```
sstv -d message.wav -o moonwalk_result.png
```

Expected output:

```
Reading WAV file...  
Decoding SSTV...  
Wrote output to: moonwalk_result.png
```

Open the resulting image:

```
xdg-open moonwalk_result.png  # Linux  
open moonwalk_result.png      # macOS  
start moonwalk_result.png      # Windows
```



The decoded image contains the flag!

Step 8: Alternative Method – QSSTV (GUI)

If you prefer a graphical interface, use **QSSTV**:

Installation (Linux):

```
sudo apt install qsstv
```

Usage:

1. Launch QSSTV
 2. Go to **File** → **Playback** → **Play**
 3. Select `message.wav`
 4. The program will automatically decode the SSTV signal
 5. The decoded image will appear in the main window
-

Step 9: Alternative Method – RxSSTV (Windows)

For Windows users:

1. Download **RxSSTV** (or MMSSTV)
 2. Open the application
 3. Click **Play** and select `message.wav`
 4. The decoded image will appear on screen
-

Step 10: Alternative Method – Online SSTV Decoder

If you don't want to install software:

1. Go to <https://ourcodeworld.com/articles/read/1041/how-to-decode-sstv-slow-scan-television-from-audio-files-using-python-in-windows>
 2. Or upload the file to an online SSTV decoder (e.g., <https://robot36.ru/>)
-



Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, the decoded SSTV image will reveal the complete flag.

Key Concepts

Concept	Description
SSTV (Slow Scan Television)	A method of transmitting images via audio frequencies, used by NASA in early space missions.
Spectrogram	A visual representation of frequencies over time; reveals hidden patterns in audio.
WAV Format	Uncompressed audio format that preserves raw signal data.
Apollo Missions	NASA used SSTV to transmit images from the moon to Earth during the Apollo program.
CMU Mascot	Carnegie Mellon University's mascot is "Scotty" the Scottish Terrier – a hint toward <code>scotty</code> mode in SSTV decoding.

Pro Tips

1. **SSTV modes vary** – There are different SSTV protocols (Martin, Scottie, Robot, etc.). Some decoders automatically detect the mode.
 2. **The CMU mascot hint** – Carnegie Mellon's mascot is "Scotty" – this points to **Scottie SSTV mode** (e.g., Scottie 1, Scottie 2).
 3. **Audacity is essential** – Before decoding, open the file in Audacity and switch to spectrogram view. If you see an image pattern, it's SSTV.
 4. **Try multiple decoders** – If one decoder doesn't work, try another (QSSTV, RxSSTV, MMSSTV, or online tools).
 5. **Check the audio duration** – SSTV images typically take 8-60 seconds to transmit. A 115-second audio file may contain multiple images.
 6. **Historical significance** – The challenge name "m00nwalk" and hint about moon landing pictures directly point to SSTV.
-

Alternative SSTV Decoders

Tool	Platform	Command/Link
sstv (Python)	Linux/macOS/Windows	<code>pip install sstv</code> or <code>git clone ...</code>
QSSTV	Linux	<code>sudo apt install qsstv</code>
RxSSTV	Windows	Download from onthesystem.com
MMSSTV	Windows	Popular SSTV decoder for amateur radio
Robot36	Android	Android app for SSTV decoding
Online Decoder	Web	https://www.sstv.de/ or https://robot36.ru/

References

- [NASA SSTV History](#)
- [SSTV Modes Explained](#)
- [Python SSTV Decoder GitHub](#)